

Chau Le Huy

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EDUCATION

National University of Singapore

Aug 2023 – Dec 2026 (Expected)

Bachelor of Computing (Honors) in Computer Science

- Specialization: Parallel Computing and Distributed System, Algorithms and Theory
- Internship availability: May 2025 – Dec 2025 (full-time credit-bearing)

ACHIEVEMENTS

- Rank 18 in International Collegiate Programming Contest (**ICPC**) – Jakarta Asia Regional (2024) (**C++**)
- Rank 15 in International Collegiate Programming Contest (**ICPC**) – Jakarta Asia Regional (2023) (**C++**)
- Second Prize in International Collegiate Programming Contest (**ICPC**) – Ho Chi Minh Asia Regional (2022) (**C++**)
- Rank 20 in Vietnam International Olympiad Team Selection Test (2022) (**C++**)
- Second Prize (Rank 24/400) in Vietnamese Olympiad in Informatics (2022) (**C++**)

EXPERIENCE

Software Engineer Intern

May 2024 – Aug 2024

Rakuna Product

- Developed a Recruiting CRM and Rakuna Scheduling Interview system using **Ruby on Rails** and **React JS**, streamlining recruitment scheduling and boosting operational efficiency by **30%**.
- Implemented multi-language support with **Rails I18n** and **I18n-JS**, enabled image/file uploads for email templates using AWS, and leveraged **Redis** caching to reduce server load and accelerate content delivery, achieving a **40% reduction in average response time**, significantly enhancing global user experience and content management.
- Built auto-reminder and process tracking features, resolved bugs, and wrote TTD tests with **RSpec**, reducing process delays and improving overall system reliability.

PROJECTS

Dokyu (dokyuu.app)

- Developed a full-stack application using **Django** and **React TypeScript** by engineering backend endpoints for uploads, email verification, and automated reminders, and by implementing robust authentication and onboarding flows. This work secured second prize at the 25th **NUS SoC Term Project Showcase** (<https://uvents.nus.edu.sg>).
- Integrated **Redis** for caching and efficient data management.
- Enhanced platform capabilities by integrating real-time video processing using **FFmpeg** in React and connecting a secure payment system via the **Stripe API**, significantly improving multimedia performance and transaction reliability.

Stellar (stellar-cs3216a3.com)

- Developed a website for analyzing personal statements for Oxbridge and Jardine applications using **generative AI**. This involved implementing a **Django** backend that handled ChatGPT API calls, securing access with token authentication and OAuth, and crafting effective AI prompts to deliver insightful and actionable feedback.
- Built key frontend components, such as upload, comment, and loading pages, using **TypeScript** with **React Vite**, which enhanced user interaction and ensured a seamless, responsive interface that significantly improved the overall user experience.

The Travel (travel-github.onrender.com)

- Developed a full-featured travel application using **ReactJS** and **ExpressJS** that includes a real-time chat system with **Socket.io** and interactive features such as destination search, booking, voting calendar, and posting destination suggestions, achieving an advanced level in the [NUS Orbital Program](#).
- Integrated third-party APIs for flight and hotel booking, significantly expanding the platform's functionality and providing users with comprehensive travel planning options.

Particle collision simulation ([GitHub](#))

- Developed a grid-based spatial partitioning algorithm using **OpenMP** for efficient broad-phase collision detection and elastic collision resolution. The simulation handles **10⁵** particles at **0.9** density over **1000-time steps** within **5 seconds**.

ADN String matching

- Developed a **CUDA**-based ADN (DNA) string matching algorithm that efficiently processed multiple pairs of signatures and samples in parallel on the GPU, leveraging spatial locality and Parallel Prefix Sum (Scan) for enhanced performance.

SKILLS

- **Languages:** Python, C++, Java, Typescript, Ruby, JavaScript, HTML/CSS
- **Frameworks and Libraries:** Django, Ruby on Rails, Express JS, Tailwind CSS, Socket.io, React
- **Tools:** OpenMP, CUDA, Redis, PostgreSQL, Stripe API, Linux, AWS SES, Google Cloud Platform, MongoDB, OpenAI API